SARAH BETTI

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SUMMARY

Highly motivated observational astronomer with an extensive background in space based near infrared and optical spectroscopy and image analysis, as well as object detection and photometry of faint astrophysical sources. Skilled in image processing, visualization, and algorithm development for visualization, calibration, and data reduction imaging pipelines. Certified Scrum Product Owner managing open source Python software for visualizing data for JWST and Roman Space Telescopes.

EDUCATION

UNIVERSITY OF MASSACHUSETTS, AMHERST — Amherst, MA
Ph.D. in Astronomy — September 2023
Advisor: Dr. Kate Follette
Dissertation: "Probing the Physical Mechanisms Responsible for Brown Dwarf and Giant Planet Formation"

UNIVERSITY OF MASSACHUSETTS, AMHERST — Amherst, MA
M.S. in Astronomy — May 2021

HAVERFORD COLLEGE — Haverford, PA
B. S. cum laude in Astrophysics — May 2017
High Honors in Astrophysics
Thesis: "VLA Observations of the Magnetic Field of the Smith High Velocity Cloud"

WORK EXPERIENCE

SPACE TELESCOPE SCIENCE INSTITUTE — Baltimore, MD
Senior Astronomical Data Scientist — April 2026 – present

- Certified Scrum Product Owner managing agile frameworks for *two* software engineering scrum teams focused on data analysis tools for the Mikulski Archive for Space Telescopes, JWST and the Roman Space Telescope. Manage product backlog, define and execute product roadmaps, and interface with stakeholders.
- Work as part of data analysis visualization tools teams to develop, build, and test open source Python and cloud-based tools. Engineered a cross-platform desktop application using Python and QT of tools, implementing notebook-based tools into custom applications for data viewing and analysis.
- Run user testing with the astronomical community to teach users, test and improve data products, and advertise tools.

NORTHROP GRUMMAN — Linthicum, MD
Principal Data Scientist — October 2025 – March 2026

- Developed python-based data processing pipelines for reading, cleaning, and visualizing device performance metrics within a jupyter notebook framework
- Build plotly-dash GUIs to visualize device-under-test performance statistics for quicklook data assessment and to make data-driven decisions on test direction and processes
- Work in cross-disciplinary team with data scientists, physicists, test engineers, and software developers serving as a point of contact for users of data analysis tools. Presented reports on tool development and analysis results to larger team with a wide range of technical comprehension.

SPACE TELESCOPE SCIENCE INSTITUTE — Baltimore, MD
STScI Prize Postdoctoral Fellow — July 2023 – October 2025

- 50% scientific research + 50% pipeline development and data analysis for the Nancy Grace Roman Space Telescope.
- Deputy Lead of the Roman Telescope Branch calibration block. Work in a cross-disciplinary team with Software Engineers, Scientists, and Analysts within the Data Calibration and Pipeline Development domains.
- Developed statistical algorithms to characterize and reduce noise by 20% in multi-TB optical imaging datasets for the Roman Wide Field Instrument (WFI). Collaborate on robust algorithmic integration and deployment of the Python-based data calibration pipeline to ensure high performance and accuracy as a member of a four-person team.
- Lead technical reports and documentation on the preprocessing algorithm for noise reduction in WFI imaging data. Presented reports to multiple stakeholders, including the NASA Roman Mission. Co-authored additional 2 technical reports on data calibration and science validation of data products.
- Conducted novel research on planet formation using multiwavelength spectroscopy from the James Webb Space Telescope. Developed advanced image processing and analysis pipeline to extract faint astrophysical signals from complex, high dimensional telescope datasets, significantly improving sensitivity in debris disk detections.

UNIVERSITY OF MASSACHUSETTS, AMHERST Amherst, MA
NASA Earth and Space Science and Technology Fellow September 2022 – July 2023

- Created CASPAR, the largest database of stellar and substellar accretion measurements. Standardized catalog entries to ensure data consistency and improved data retrieval resulting in 17% reduction in object variability. Implemented customized linear regression models using pandas and scikit-learn machine learning tools to analyze statistical trends resulting in scientific publication (Betti+2023).
- Built Python-based data reduction pipelines for flux calibration and data visualization to analyze near infrared spectroscopy of brown dwarf stars, detecting features at the 20% level, and resulting in scientific publication (Betti+2022b).

UNIVERSITY OF MASSACHUSETTS, AMHERST Amherst, MA
Graduate Research and Teaching Assistant September 2017 – September 2022

- Developed image reduction pipeline, performed flux calibration, data analysis, and simulation modeling to visualize and robustly detect high contrast and high resolution imaging and spectroscopy of circumstellar disk environments, resulting in publication (Betti+2022a). Designed Python-based algorithm to improve distortion and amplifier offsets in image data.
- Built full reduction and analysis pipeline from IDL to Python for the TolTEC Clouds To Cores Legacy Survey for the Large Millimeter Telescope to predict object detection and feature extraction using k-nearest-neighbors and watershed image segmentation in simulated datasets, resulting in scientific publication (Betti+2021).
- Assisted teaching and mentoring research based observational astronomy courses. Observed at high altitude sites, such as the 0.9 m WIYN telescope at Kitt Peak Observatory. Commissioned Amherst College 12" Schmidt-Cassegrain telescope to image guiding and focus.

HAVERFORD COLLEGE Haverford, PA
Thesis Research Writer June 2017
Undergraduate Research Assistant September 2015 – May 2017

NATIONAL RADIO ASTRONOMY OBSERVATORY Green Bank, WV
Research Experience for Undergraduates Intern 2015

CERTIFICATIONS

Certified Scrum Product Owner, Scrum Alliance, 2026

PROFESSIONAL DEVELOPMENT AND TECHNICAL SKILLS

Machine Learning/Comprehensive Python for Data Science, Accelebrate, November 2024
Programming languages: Python, IDL, LaTeX, (basic) SQL
Machine Learning Libraries: scikit-learn
Version control: GitHub/GitLab/Grit
Software applications: Confluence, JIRA, Microsoft Office Suite
Data Visualization and Analysis Tools: Jupyter Notebook, NumPy, Pandas, matplotlib, CASA, SAOImage ds9, yt, pyKLIP, MCFOST, photutils, specutils, Jdaviz
Telescope data reduction pipeline development (Roman, LMT, LBT/LMIRCam)
Multiwavelength Data Reduction Experience:

- Optical Imaging: WIYN 0.9 m telescope
- IR Imaging: LBT, APO, IRTF, JWST
- IR Spectroscopy: SOAR
- Sub-mm Interferometry imaging: Atacama Large Millimeter Array
- Sub-mm imaging: Large Millimeter Telescope Alfonso Serrano
- Radio Interferometry imaging : Very Large Array

Observing Experience:

- Keck I/LRIS
- Southern Astrophysical Research Telescope/TripleSpec 4.1
- Apache Point Observatory/TripleSpec
- IRTF/iSHELL
- Kitt Peak Observatory 0.9 m WIYN telescope/HDI
- 11” al-az and equatorial Schmidt-Cassegrain Amherst College telescopes including the SBIG CCD

OPEN SOURCE SOFTWARE and DATABASES

Leadership Roles

- **Product Owner**: STScI agile software engineering scrum team, developing Jdaviz, a data visualization and analysis tools for JWST and Roman telescopes (2026-present)
- **Product Owner**: STScI agile software engineering scrum team, developing Python-based widgets for cloud-based platforms for data visualization of Roman Space telescope data (2026-present)
- **Owner/Creator**: created the Comprehensive Archive of Substellar and Planetary Accretion Rates, the largest astronomical archive of accretion rates for stars, brown dwarfs, and planets.

Projects

- **Maintainer** of JDAT notebooks, a repository of Python jupyter notebooks for data analysis of JWST observations (2026-present)
- **Creator** of the Comprehensive Archive of Substellar and Planetary Accretion Rates
- **Creator and maintainer** of casparing, an open source Python package for analysis and plotting tools for use with CASPAR (2025-present)
- **Creator and maintainer** of bdfit, an open source Python package for spectral fitting routines for low resolution brown dwarf near-infrared spectra (2025-present)
- **Creator** of ImSeg, an open source Python package for data reduction and analysis of LMT TolTEC Clouds2Cores survey

Workshops and Webinars

- **Leader**: Virtual Webinar, "Hands-on Demo, Viewing, and Analyzing JWST Data with Jdaviz 5.0" July 2026

REFEREED PUBLICATIONS: FIRST AUTHOR

Betti, S. K., et al., "The Comprehensive Archive of Substellar and Planetary Accretion Rates", AJ, 166, 262B, 2023.

Betti, S. K., "Probing the Physical Mechanisms Responsible for Brown Dwarf and Giant Planet Formation", PhD Thesis, University of Massachusetts Amherst, 2023.

Betti, S. K., et al., "Erratum: Near-infrared Accretion Signatures from the Circumbinary Planetary Mass Companion Delorme 1 (AB)b", ApJL, 941, L20, 2022.

Betti, S. K., et al., "Near-infrared Accretion Signatures from the Circumbinary Planetary Mass Companion Delorme 1 (AB)b", ApJL, 935, L18, 2022.

Betti, S. K., et al., "Detection of Near-Infrared Water Ice at the Surface of the (pre)Transitional Disk of AB Aur: Informing Icy Grain Abundance, Composition, and Size", AJ, 163, 145, 2022.

Betti, S. K., et al., "The Robustness of Synthetic Observations in Producing Observed Core Properties: Predictions for the TolTEC Clouds to Cores Legacy Survey", ApJ, 923, 25, 2021.

Betti, S. K., et al., "Environmental Effect on the Interstellar Medium in Galaxies across the Cosmic Web at z = 0.73", ApJ, 874, 53, 2019.

Betti, S. K., et al., "Constraining the Magnetic Field of the Smith High-velocity Cloud Using Faraday Rotation", ApJ, 871, 215, 2019.

NON-REFEREED PUBLICATIONS: FIRST AUTHOR

Technical Reports

Betti, S. K., et al., "The Statistical Properties of Dark Ramps for the Roman-WFI Detectors, Technical Report Roman-STScI-000814, 2025.

Betti, S. K., et al., "Application and Characterization of IRRC on Roman/WFI TVAC1 Data", Technical Report Roman-STScI-000673, 2024.

REFEREED PUBLICATIONS: CO-AUTHOR

Mathilde, M, et al., incl. **Betti, S. K.**, "JWST-TST High Contrast: Medium-resolution spectroscopy reveals a carbon-rich circumplanetary disk around the young accreting exoplanet Delorme 1 AB b", Astronomy & Astrophysics, 704, A181, 2025.

Worthen, K., et al., incl. **Betti, S. K.** "Fluorescently excited CO emission in the 49 Ceti debris disk spatially resolved by JWST/NIRSpec", Nature Astronomy, 9, 11, 2025.

Manjavacas, E., Gennaro M., **Betti, S. K.**, "A Near-infrared Spectral Library of Very Young Brown Dwarf and Planetary Mass Objects in the Orion Nebula Cluster", The Astrophysical Journal, 994, 1, 2025.

Xie, C., et al., incl. **Betti, S. K.**, "Water ice in the debris disk around HD 181327", Nature, 641, 8063, 2025.

Ray, S., et al., incl. **Betti, S. K.**, "The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems III: Aperture Masking Interferometric Observations of the star HIP 65426 at 3.8μm", ApJ, 983L, 25R, 2025.

Rebollido, I., et al., incl. **Betti, S. K.**, "JWST-TST High Contrast: Asymmetries, dust populations and hints of a collision in the β Pictoris disk with NIRCam and MIRI", AJ, 167, 69, 2024.

Sallum, S., et al., incl. **Betti, S. K.**, "The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems IV: NIRISS Aperture Masking Interferometry Performance and Lessons Learned", ApJ, 963L, 2S, 2024.

Jorquera, S., Bonnefoy, M., **Betti, S. K.**, et al., "LBTI search for companions and sub-structures in the (pre)transitional disk of AB Aurigae", ApJ, 926, 71, 2022.

Hutschenreuter, S., Anderson, C. S., **Betti, S. K.**, et al., "The Galactic Faraday rotation sky 2020", A&A, 657, A43, 2022.

Pokhrel R., Gutermuth R. A., **Betti S. K.**, et al., "Star-Gas Surface Density Correlations in Twelve Nearby Molecular Clouds I: Data Collection and Star-Sampled Analysis", ApJ, 896, 1, 2020.

Ma Y. K., et al., incl. **Betti, S. K.**, "A broad-band spectro-polarimetric view of the NVSS rotation measure catalogue - II. Effects of off-axis instrumental polarization", MNRAS, 487, 3, 2019.

Ma Y. K., et al., incl. **Betti, S. K.**, "A broad-band spectro-polarimetric view of the NVSS rotation measure catalogue - I. Breaking the nπ-ambiguity", MNRAS,487, 3, 2019.

NON-REFEREED PUBLICATIONS: CO-AUTHOR

Chen, C., Xie, C., **Betti, S. K.**, et al. "Detecting and Characterizing Volatile Ices in Extrasolar Kuiper Belts", PRIMA General Observer Science Book Volume 2, Ed., 2025.

Dacus, B., et al., incl. **Betti, S. K.**, "Toward Assembling a Comprehensive Database of Substellar Accretion Rates", RNAAS, 5, 174, 2021.

Lockman F., **Betti S. K.**, et al.,"High velocity Clouds: Building Blocks of the Local Group?" Astro2020: Decadal Survey on Astronomy and Astrophysics, science white papers, no. 255; Bulletin of the American Astronomical Society, Vol. 51, Issue 3, id. 255, 2019.

Technical Reports

Desjardins, T. D. **Betti. S. K.**, et al., "Roman SOC Science Validation Report for SOC Release 2 DMS", Technical Report Roman-STScI-000619, 2024.

Petric, A. et al., incl. **Betti, S. K.**, "Approximating Uncertainties for Calibration Reference Files", Technical Report Roman-STScI-000571, 2024.

Conference Proceedings

Ma Y. K., et al., incl. **Betti, S. K.**, "From the NVSS RM Catalogue to Future Polarisation Surveys", Proceedings to IAU Focus Meeting: New Insights in Extragalactic Magnetic Fields, XXXth IAU General Assembly, 2018.

Kobelski A., Bastian T. S., **Betti, S. K.**, "Probing Solar Wind Turbulence with the Jansky Very Large Array", Coimbra Solar Physics Meeting: Ground-based Solar Observations in the Space Instrumentation Era Proceedings of a Meeting (eds.) Ivan Dorotovic, Catherine E. Fischer, and Manuela Temmer, (San Francisco: ASP), 504, 2015.

PRESENTATIONS & WORKSHOPS

Talks

National Capital Area Disks Meeting #8, Carnegie Science Earth and Planets Laboratory, Washington, DC — March 2025
"Detection of Icy Volatiles in β Pictoris using JWST/NIRSpec"

Star and Planet Formation Seminar, Space Telescope Science Institute, Baltimore, MD — August 2024
"Water Ice in the Debris Disk of β Pictoris"

Star and Planet Formation Seminar, ESO Headquarters, Garching, Germany — February 2024
"CASPAR: Characterizing Accretion across the Brown Dwarf Regime'"

Planet and Star Formation Tea, MPIA, Heidelberg, Germany — December 2023
"CASPAR: Characterizing Accretion across the Brown Dwarf Regime"

American Astronomical Society Winter Meeting, Seattle, WA — January 2023
"Characterizing Accretion and Formation Mechanisms across the Brown Dwarf and Planetary Mass Regimes"

Planet and Star Formation Tea, MPIA, Heidelberg, Germany — November 2022
"NIR Accretion Signatures in the Circumbinary Planetary Mass Companion Delorme 1 (AB)b"

Planetary Lunch Series, Massachusetts Institute of Technology, MA — October 2022
"Observing Water Ice in the AB Aurigae Transitional Disk"

Northeast Star and Planet Formation 2022, Wesleyan University, CT — July 2022
"The Role of Environment in Core Evolution"

Cool Stars 21: Gaia and Spitzer's Neighbourhood Watch Program: From Stars to Brown Dwarfs to Planets Splinter Session, Toulouse, France — July 2022
"NIR Accretion Signatures in the Circumbinary Planetary Mass Companion Delorme 1 (AB)b"

In the Spirit of Lyot 2022, Leiden, Netherlands — June 2022
"Mapping Icy Grains in the Circumstellar Disk around AB Aurigae"

Five Years after HL Tau: A New Era in Planet Formation Conference, Virtual — December 2020
"Mapping Icy Grains in the Circumstellar Disk around AB Aurigae"

Southern Maine Astronomy Club Public Lecture, Virtual — November 2020
"Formation of Exoplanets in Circumstellar Disks"

Past, Current and Future Galaxy Surveys. CANDELS Meeting and TolTEC Workshop, University of Massachusetts, Amherst, MA, — October 2018
"Environmental Effect on Interstellar Medium in Galaxies across the Cosmic Web at z=0.73"

Physics with Friends, Haverford, PA — November 2016
"Gas Clouds in Space"

National Radio Astronomy Observatory Lunch Talk, Jansky Lab, Green Bank, WV — July 2015
"Type III Bursts and the Solar Wind"

Posters

Cool Stars 22, San Diego, CA — July 2024
"Characterizing Accretion and Formation Mechanisms across the Brown Dwarf and Planetary Mass Regimes"

Extreme Solar Systems V, Christchurch, NZ — March 2024
"Coronagraphic JWST/NIRCam Images of the 49 Ceti Debris Disk"
Origins of Solar Systems Gordon Research Seminar, Holyoke, MA — June 2023
"Detection of Near-infrared Water Ice at the Surface of the (Pre)Transitional Disk of AB Aur"
Planetary Systems and the Origins of Life in the Era of JWST, Baltimore, MD — May 2023
"Characterizing Accretion and Formation Mechanisms across the Brown Dwarf and Planetary Mass Regimes"
Protostars and Planets VII, Kyoto, Japan — April 2023
"Characterising Accretion and Formation Mechanisms across the Brown Dwarf and Planetary Mass Regimes"
Cool Stars 21, Toulouse, France — July 2022
"NIR Accretion Signatures in the Circumbinary Planetary Mass Companion Delorme 1 (AB)b"
2021 Sagan Exoplanet Summer Virtual Workshop: Circumstellar Disks and Young Planet, Virtual — July 2021
"Mapping Icy Grains in the Transitional Disk AB Aur"
2021 Virtual STScI Spring Symposium: Towards the Comprehensive Characterization of Exoplanets: Science at the Interface of Multiple Measurement Techniques, Virtual — April 2021
"Mapping Icy Grains in the Transitional Disk AB Aur"
American Astronomical Society Winter Meeting, Virtual — January 2021
"Mapping Icy Grains in AB Aur: Constraining Composition, Growth, and Filtration"
American Astronomical Society Winter Meeting, Honolulu, HI — January 2020
'The Role of Environment in Core Formation: Predictions for the TolTEC Clouds to Cores Legacy Survey"
American Astronomical Society Winter Meeting, National Harbor, Oxon Hill MD — January 2018
"Quantifying the Role of Environment in Star Formation: ISM masses along the Cosmic Web with ALMA"
American Astronomical Society Winter Meeting, Grapevine, TX — January 2017
"VLA Observations of the Magnetic Field of the Smith High Velocity Cloud"
Start Talking Science, Chemical Heritage Foundation, Philadelphia, PA — August 2016
"Gas Clouds in Space: How the Smith Gas Cloud survived passing through the Milky Way"
Koshland Natural Integrated Science Center Research Symposium, Haverford College, Haverford PA — August 2016
"Gas Clouds in Space: How the Smith Gas Cloud survived passing through the Milky Way"
American Astronomical Society Winter Meeting, Kissimmee, FL — January 2016
"Probing Solar Wind Turbulence Using JVAS and VLA Calibrator Sources"

Panels

Young Women in Physics Conference, Bucknell University, Lewisburg, PA — April 2016

Workshops

Circumstellar Disks and Young Planets, 2021 Sagan Exoplanet Summer Virtual Workshop, Virtual — July 2021
ALMA Proposal Workshop, Amherst, MA — March 2020
JWST Proposal Planning Workshop, Amherst, MA — January 2020
16th NRAO Synthesis Imaging Workshop, Soccoro, NM — May 2018

FELLOWSHIP AND GRANTS

James Webb Space Telescope Cycle 4 General Observer | $130,430 — 2025
STScI Prize Postdoctoral Fellowship, Space Telescope Science Institute | $32,000 — 2023-2027
NASA Earth and Space Science and Technology Fellowship | $50,000 — 2022-2023
UMass Amherst Department of Astronomy Research Grant | $500 — 2022
UMass Amherst Department of Astronomy Research Grant | $100 — 2021
UMass Graduate School Dissertation Research Grant | $750 — 2020
Massachusetts Space Grant Consortium Fellowship | $5500 — 2019
Mary Dailey Irvine Graduate Travel Award | $800 (2018), $1192 (2019), $900 (2022) — 2018-2022
Massachusetts Space Grant Consortium Fellowship | $5500 — 2018
Frances Velay Womens' Science Research Fellowship — 2016

SELECT OBSERVATIONAL PROGRAMS (*PI)

JWST:

- *13.7 hours (PID 6940, PI: S. Betti)
- 7.2 hours (PID 7313, PI: K. Worthen)
- 17.29 hours (PID 6086, PI: K. Ward-Duong)
- 20.92 hours (PID 6361, PI: K. Follette)
- 4.11 hours (PID 5390, PI: C. Lu)

Keck I:

- 1 night (2022B, PI: K. Follette)
- 1.5 nights (2021B, PI: K. Follette)
- 1 night (2021A, PI: K. Follette)

Hubble Space Telescope:

- 9 orbits (PID 17122, PI: C. Robinson)
- 5 orbits (PID 18026, PI: C. Lu)

SOAR 4.1m:

- 0.5 nights (PID 2023B-190386, PI: C. Robinson)
- *2 nights (PID 2023A-267101, PI: S. Betti)
- *3 nights (PID 2022B-171130, PI: S. Betti)
- *2 nights (PID 2022A-336353, PI: S. Betti)
- *4.5 nights (PID 2021B-0311, PI: S. Betti)
- *1 night (PID 2021A-0261, PI: S. Betti)

VLT:

- 3.0 hours (116.290W, PI: C. Chen)
- 3.6 hours (115.29GA, PI: D. Demars)
- 6 hours (113.26AR, PI: S. Ringqvist)
- 7.5 hours (113.26J1, PI: D. Demars)

ALMA:

- 4 runs (2017.1.01347.S, PI: A. Pope)
- 3 runs (2017.1.00091.S, PI: A. Pope)

APO 4.5m:

- *1 night (2024 JH03, PI: S. Betti)

TEACHING EXPERIENCE

UMass Precollege Summer Astronomy Program 2019–2021

Course Head and Coordinator | 2020–2021 *Amherst, MA*

Organized and ran a 3 week pre-college astronomy course centered around students learning fundamentals of astronomy through lectures and optical data reduction labs; supervised graduate student teachers, ran observing nights, and organized final student presentations.

Lab Coordinator | 2019, 2022

Developed and taught 3-week Python based optical data reduction and analysis lab course. Organized and managed an observing night at the Amherst College Observatory using 11" Schmidt-Cassegrain telescopes with eyepieces and CCDs.

University of Massachusetts, Amherst 2018–2021

F*CAD Graduate Observational Teaching Assistant | Spring 2018–Fall 2021* *Amherst, MA*

Astronomy 337 & 341 Observational Astronomy courses: Research based courses using the Smith College 12" and 16" Schmidt-Cassegrain telescope and the 0.9 m telescope at Kitt Peak Observatory. Assisted in teaching & mentoring students in conducting research projects.

Lab Teaching Assistant | Fall 2017

Ran and taught Astronomy 100 Laboratory and Discussion sections. Introduced students to basic astronomy concepts using Stellarium.

Haverford College 2015–2017

Telescope/Lecture Teaching Assistant | 2015–2017 *Haverford, PA*

Trained Astronomy 205: Intro to Astrophysics students on 12" Schmidt-Cassegrain telescope. Assisted Astronomy 206: Intro to Astrophysics students on problem sets and coding in Python.

Grader | 2015–2016

Graded Fundamental Physics student problem sets (Fall 2015), Introduction to Electrodynamics student problem sets (Spring 2016), and Waves and Optics student problem sets (Fall 2016), Introduction to Astrophysics student problem sets (Spring 2017)

OUTREACH, SERVICE & ACTIVITIES

Reviewer for: ApJ

Maryland Science Center 2024-2025

Observatory Volunteer *Baltimore, MD*

Ran safe solar observing at the Clark Observatory. Set up and operate 8" aperture refractor telescope and dome, engage the public through discussions of solar sunpots, telescope filters, and general astronomy. Ran astronomy-based demonstrations, such as demonstrating the physics of black holes and spectroscopy.. Operate the Science on a Sphere visualization system to showcase and describe solar system bodies in 3D.

Telescope Allocation Committee Member for NOIRLab 2024

Space Telescope Science Institute 2024-2025

Diversity, Culture, Respect Working Group | Feb. 2024–Jan. 2025 *Baltimore, MD*

Part of the DCRWG whose mission is to analyze and address issues of diversity, culture and respect within the STScI Instruments Division. WG disbanded as a result of the executive order issued on January 20, 2025.

JWST Cycle 3 Panel Support Scientist | Feb. 2024

ULLYSES: Continuing the Voyage of Discovery Workshop at STScI March 2024

Science Organizing Committee, Conference Moderator Baltimore, MD

University of Massachusetts, Amherst 2019–2023

UMass Amherst Graduate Student Senate Senator | 2020–2023 *Amherst, MA*

Graduate Women in Stem Sound Bites Cafe Volunteer | 2020–2021

Led discussion and gave presentations about exoplanets and their formation to middle and high school science classes

Department of Astronomy, University of Massachusetts, Amherst 2018–2023

Member of Graduate DEI Committee | 2020–2023 *Amherst, MA*

Grad student-led committee formed to promote DEI within the astronomy department including retention, diversity, and inclusion within all levels of the department.

Faculty Meeting Graduate Student Representative | Fall 2019, Fall 2020

Graduate Student Recruitment Committee | 2018–2022

Astronomy in the Community, University of Massachusetts, Amherst 2018–2020

Interview with an Astronomer | 2020 *Amherst, MA*

Led discussions about social issues in space science and was interviewed for a sixth grade project

Astronomy Day Girl Scout Camp Volunteer | Summer 2018

Ran day long event teaching and leading astronomy activities to kindergarten to middle school girl scouts

Local High School Astronomy Club | Spring 2018

Led discussions and presentations about hot astronomy topics including gravitational waves and black holes for students in high school astronomy

Astronomy Public Observing Program, Haverford College 2014–2017

Astronomy Public Outreach Head | 2015–2017 *Haverford, PA*

Managed and organized the public observing program. Interacted with professional astronomers and astronomy. Delegated jobs and oversaw volunteers at events. Hosted private groups for personal observing by giving talks, leading discussions, and running telescopes.

Public talk: "The Sun: From Birth to Death", March 18, 2016

Astronomy Public Outreach Volunteer | 2014–2015

Ran telescopes and helped with crafts at public observing program

Public talk: "Constellations: What are they and Their Place in Modern Astronomy", March 21, 2015

Haverford College 2014–2017

Koshland Integrated Natural Science Center Student Advisory Committee *Haverford, PA*

Met with the KINSC program coordinator and the faculty to ensure student funding and programs were available for all Haverford students

ORGANIZATION MEMBERSHIP

2016-present: American Astronomical Society
2011-2019: International Dark Sky Association
2016-2017: Society of Physics Students